



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
REVIEW TEST
27-02-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TOAS, TNAS & TAAS]

Time : 3 Hours

PAPER-2

Maximum Marks : 198

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS**BEWARE OF NEGATIVE MARKING****PART-1 : PHYSICS****SECTION-I : (Maximum Marks: 24)**

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

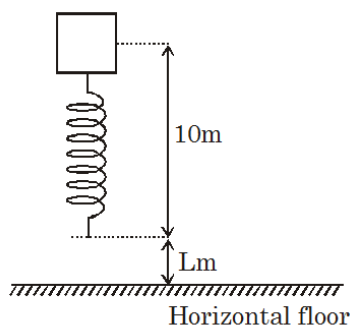
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

Space for Rough Work

1. A small block of mass m is fixed at upper end of a massless vertical spring of spring constant $K = \frac{4mg}{L}$ and natural length 10 m . The lower end of spring is free and is at a height L meter from fixed horizontal floor as shown. The spring is initially unstretched and the spring block system is released from rest in the shown position.

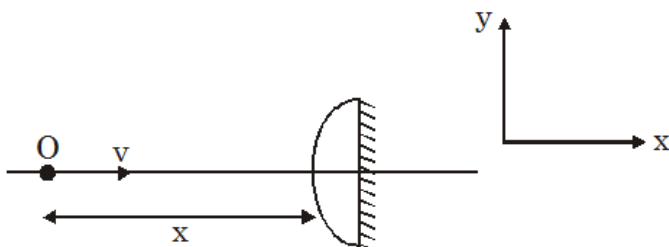


- (A) When the speed of block is maximum the magnitude of force exerted by spring on the block is mg .
 (B) The maximum speed attained by the block is $\frac{3}{2}\sqrt{gL}$.
 (C) The maximum compression of the spring is $\frac{L}{8}$

The time taken by the mass to move from the equilibrium position to the maximum compressed

- (D) position of the spring is $\frac{\pi}{4}\sqrt{\frac{L}{g}}$

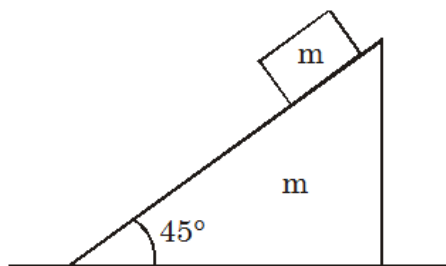
2. When object 'O' moves towards a fixed lens mirror combination, select correct choice/choices:



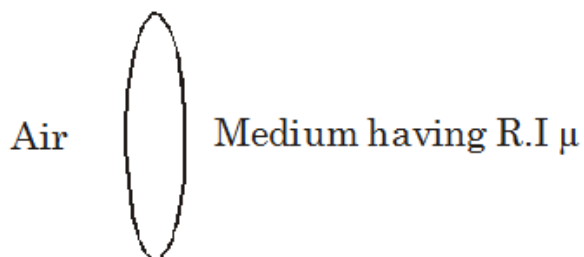
- (A) Image moves towards negative x -axis. (B) Speed of image and object may be same
 (C) Image may move faster than object (D) Image may come closer to arrangement

Space for Rough Work

3. A block having mass m is placed over a inclined plane of same mass and inclination 45° if coefficient of friction between all surface is same it is μ . Now system set free to move then



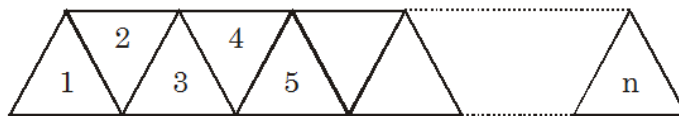
- (A) For $\mu > 1$ block over incline will move but wedge don't
 (B) For $0.2 < \mu < 1$ block will move but wedge will not
 (C) For $\mu = 0.15$ both will move
 (D) It is not possible to move wedge without moving block over wedge
4. An equiconvex lens of focal length 20 cm having a glass of refractive index $\frac{3}{2}$ is used in a medium air on left side and another medium having refractive index μ on the right side of lens as shown in figure.



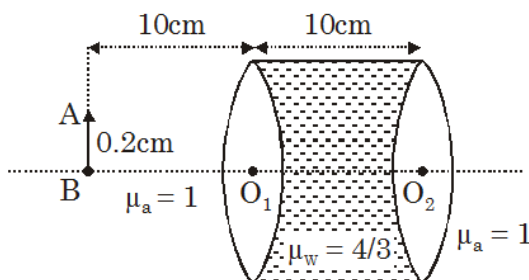
- (A) Character of lens will be diverging for $\mu > 2$
 (B) Focal length of lens will be 40 cm for $\mu = 4$ and character will be diverging
 (C) Object distance and image distance with respect to lens will be same for $\mu = 2$
 (D) Character of lens can't be explained

Space for Rough Work

5. n number of identical equilateral prisms are kept in contact as shown in figure. If deviation through a single prism is δ , then (n, m are integers)



- (A) if $n = 2m$, deviation through n prism is zero.
 (B) if $n = 2m + 1$, deviation through system of n prisms is δ .
 (C) if $n = 2m$, deviation through system of n prism is δ .
 (D) if $n = 2m + 1$ deviation through system of n prisms is zero
6. Consider an arrangement of two thin equiconvex lenses made of glass ($\mu_g = 3/2$) and each of focal length in air 10 cm are separated by a distance 10 cm. The space between the two lenses is filled with water of refractive index $\mu_w = \frac{4}{3}$. A small linear object 'AB' of size 0.20 cm is placed perpendicular to the optic axis at a distance 10 cm from the first lens in air as shown. Then choose the correct option(s) :



- (A) The position of the final image formed is 25 cm right to the second lens.
 (B) The position of the final image formed is 40 cm right to the second lens.
 (C) The size of the final image formed is 0.40 cm.
 (D) The size of the final image formed is 0.60 cm.

Space for Rough Work

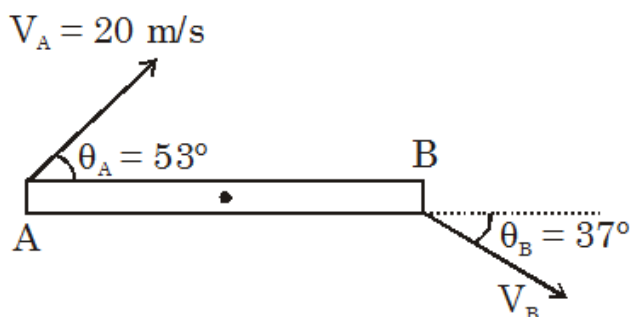
SECTION-II : (Maximum Marks: 24)

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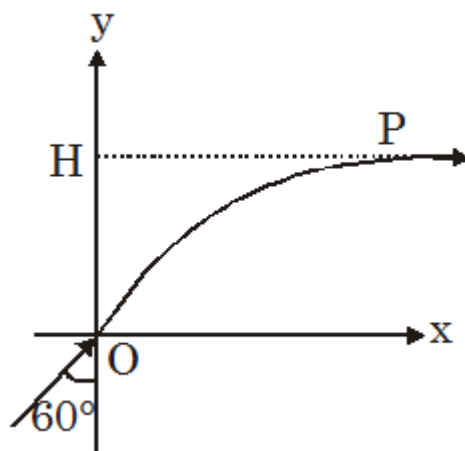
Zero Marks : 0 In all other cases.

1. A projectile is thrown with a velocity of 18 m/s at an angle of 60° with horizontal. The interval between the moments (in sec) when speed of the projectile is 15 m/s is ($g = 10 \text{ m/s}^2$)
2. The two ends A and B of a uniform rod of length $\ell = 1\text{m}$ are moving with velocities V_A and V_B as shown. The velocity V_B (m/s) is



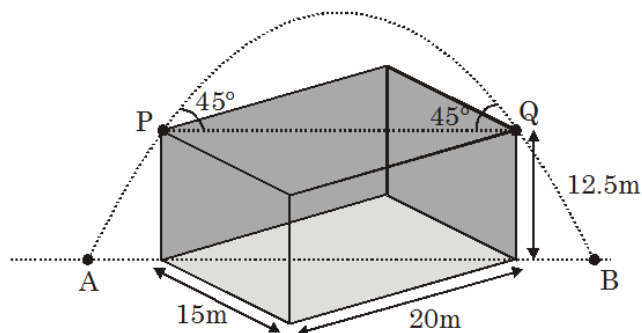
Space for Rough Work

3. A system of coordinates is drawn in a medium whose refractive index varies $\mu = \frac{2}{1+y^2}$ as, where $0 \leq y \leq 1$. A ray of light is incident at origin at an angle 60° with y-axis as shown in the figure. At point P ray becomes parallel to x-axis. Find the value of 'H'. (Take $\mu = 2$ for $y < 0$)

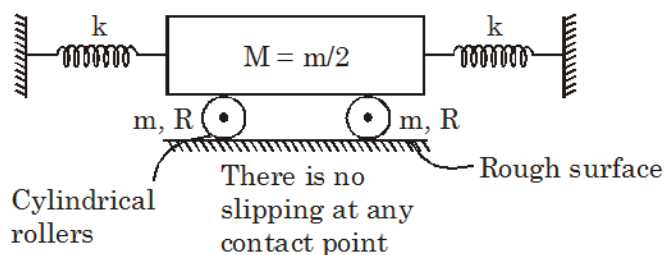


Space for Rough Work

4. A commander fires a shell at certain angle of projection from 'A' which clears the building (cuboid) of dimensions $20 \times 15 \times 12.5 \text{ m}^3$ in its diagonal plane. Shell just clears the roof diagonally at points P and Q and falls on the other side of the building at B, and makes 45° with horizontal at P and Q as shown in the figure. If the range AB of the shell is $N \times \sqrt{3}$ metre then find N. ($g = 10 \text{ m/s}^2$)

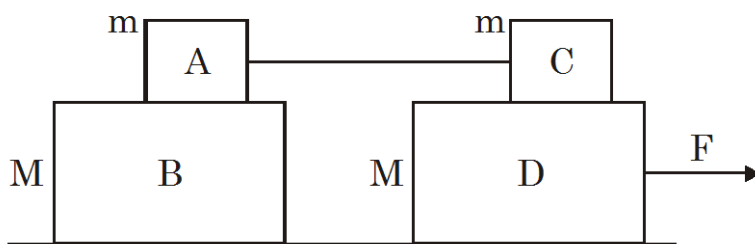


5. In the figure shown below all the springs are ideal and time period of small oscillations about equilibrium position is $\pi \sqrt{\frac{xm}{k}}$. Find the value of x.



Space for Rough Work

6. Four blocks are arranged on a smooth horizontal surface as shown. The masses of the blocks are given (see the figure). The coefficient of static friction between the top and the bottom blocks is μ_s . What is the maximum value of the horizontal force F , applied to one of the bottom blocks as shown, that makes all the four blocks move with the same acceleration? [Take : $\mu_s = 0.5$, $g = 10 \text{ m/s}^2$, $m = 1 \text{ kg}$, $M = 2 \text{ kg}$]



Space for Rough Work

SECTION-III : (Maximum Marks: 18)

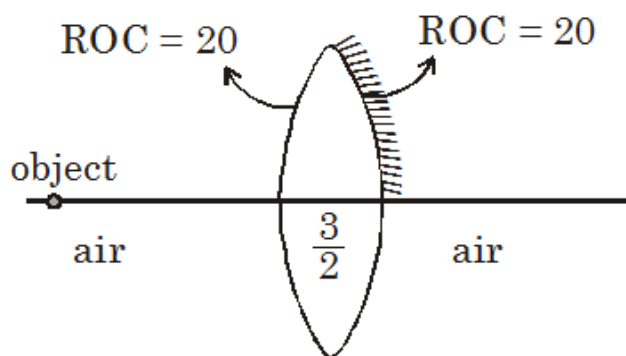
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Zero Marks : 0 If no answer is given.

Negative Marks : -1 In all other cases

1. Distance between two images formed by upper and lower part of the point object placed at 30 cm from given lens is $(60 + x)$ cm, then the value of x is :



2. A block of mass 2 kg is gently placed over a massive plank moving horizontally over a smooth surface with velocity 6 m/s. The coefficient of friction between the block and plank is 0.2. The distance travelled by the block till it slides on the plank is ($g = 10 \text{ m/s}^2$)

Space for Rough Work

3. The minimum deviation suffered by a ray emerging through a prism $\left(\mu = \frac{2}{\sqrt{3}}\right)$ is equal to half the refracting angle of the prism. Find the value of C if the refracting angle of the prism is $\left(\frac{2\pi}{C}\right)$.
4. It is desired to photograph the image of an object placed at a distance of 3m from the plane mirror. The camera which is at a distance of 4m from the mirror should be focussed for a distance of (in m)
5. The potential energy of a particle of mass 1 kg in motion along the x-axis is given by $U = (10 - 10 \cos 6x)$ J. If the period of small oscillations is $\frac{1}{n}$ second, find the value of n . ($\pi^2 \approx 10$) .
6. When a plane mirror is placed horizontally on level ground at a distance of 9 m from the foot of a tower, the ray coming from the top of the tower and its reflected ray from the mirror subtends an angle of 90° . What is the height of the tower ?

Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I : (Maximum Marks: 24)

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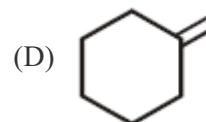
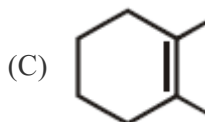
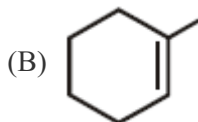
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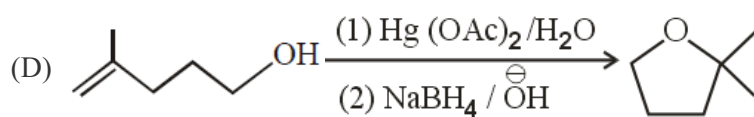
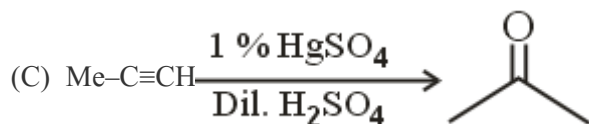
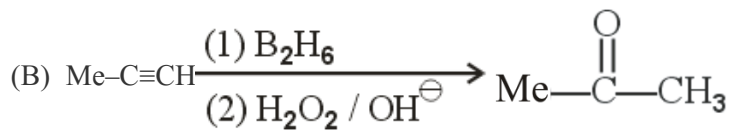
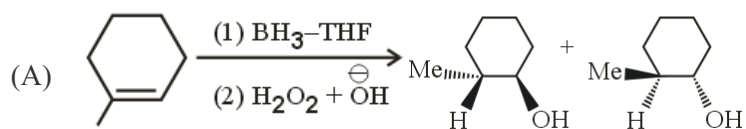
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1. Markovnikoff's rule is applicable on which of the following alkene :



Space for Rough Work

2. Identify reactions correctly matched with their major products ?

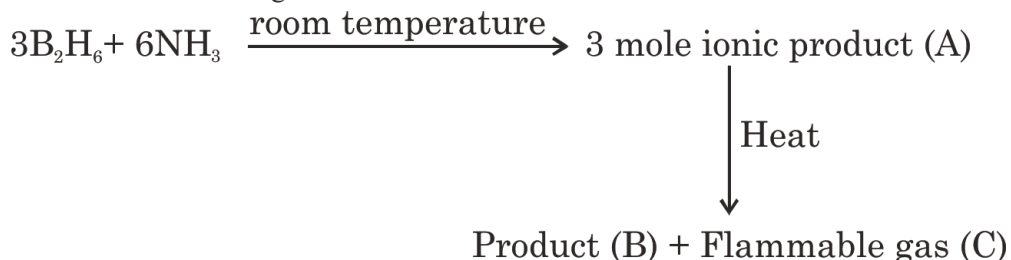


Space for Rough Work

3. Select correct statement(s)
- (A) Number of second nearest neighbours in primitive cubic unit cell is 12.
- (B) Minimum inter-ionic distance in Na_2O is $\sqrt{3} a/2$.
- (C) Distance between two consecutive hexagonal layer in FCC is $4r \sqrt{2/3}$.
- (D) Number of next nearest neighbours for S^{2-} in ZnS type structure is 12.
4. At 40°C , vapour pressure in Torr of methanol and ethanol solution is $P = 119x + 135$ where x is the mole fraction of methanol. Hence
- (A) vapour pressure of pure methanol is 254 Torr
- (B) vapour pressure of pure ethanol is 135 Torr
- (C) vapour pressure of equimolar mixture of each is 194.5 Torr
- (D) vapour pressure of pure methanol is 119 Torr

Space for Rough Work

5. Lone pair donating power of $(\text{H}_3\text{C})\text{N}(\text{SiH}_3)_2$ molecule is enhanced by :-
- (A) Replacing only $-\text{CH}_3$ by $-\text{SiH}_3$ (B) Replacing one $-\text{SiH}_3$ by $-\text{CH}_3$
- (C) Replacing both $-\text{SiH}_3$ by $-\text{CH}_3$ (D) Replacing both $-\text{SiH}_3$ by $-\text{Hydrogen}$
6. Consider the following reaction and choose correct statements



- (A) Product (B) contain aromatic character.
- (B) Boiling point of product (C) is less than Helium.
- (C) All Boron in product (A) are sp^3 hybridised.
- (D) In product (B), $2p_\pi-3p_\pi$ type back bond present.

Space for Rough Work

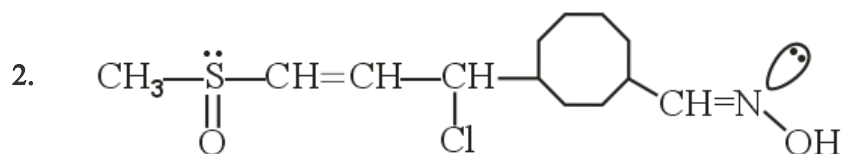
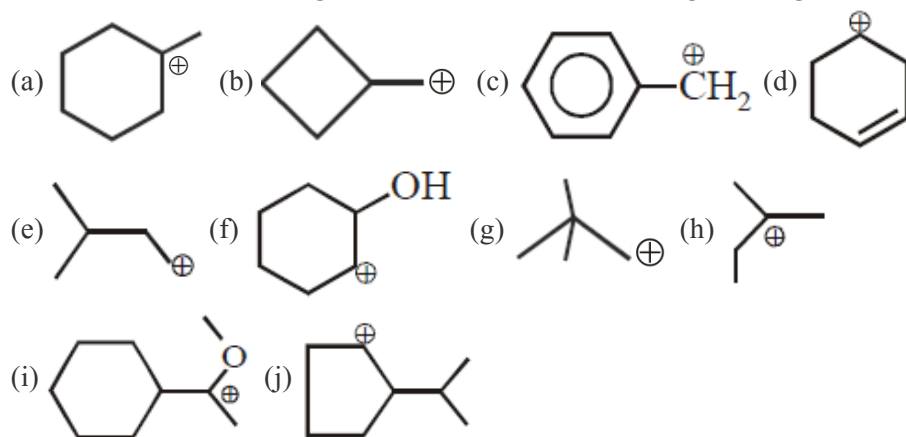
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1. Total number of following carbocation which will undergo rearrangement ?

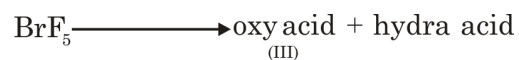
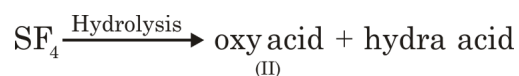
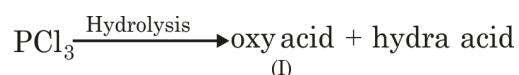


If total number of possible stereoisomers for above molecule are 2^n .

The value of n will be :

Space for Rough Work

3. The normal boiling point of pure water is 373.15 K. Calculate the boiling point (in K) of an aqueous solution containing 11.7 gm NaCl and 34.2 gm sugar ($C_{12}H_{22}O_{11}$) in 260 gm water. For water, $K_b = 0.52 \text{ K-Kg/mol}$. [Atomic mass of Na = 23, Cl = 35.5]
4. Henry's law constant for CO_2 in water is $1.6 \times 10^8 \text{ Pa}$ at 298K. The mass of CO_2 (in gm) is 500g of soda water when packed under 3.2 bar pressure at 298K.
5. In which of the following molecule $p\pi - d\pi$ back bonding present?
 $CHCl_3$, $B_3N_3H_6$, $(CH_3)_3N$, SiH_3NCO , $:CCl_2$, $(CH_3)_3SiOH$, $(SiH_3)_2O$, $:CF_2$
6. Write sum of basicity of oxy acid formed by hydrolysis of :-



Space for Rough Work

SECTION-III : (Maximum Marks: 18)

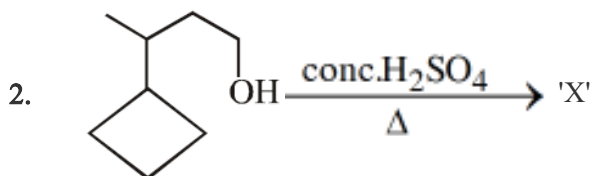
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1. How many product(s) is/are formed when trans but-2-ene reacts with $\text{Br}_2 / \text{CCl}_4$



Total number of 1,2 shifts (rearrangements) involve in the above reaction.

3. In an ionic solid $r_{(+)} = 1.6 \text{ \AA}$ and $r_{(-)} = 1.864 \text{ \AA}$. Use the radius ratio rule to determine the edge length of the cubic unit cell in $\text{\AA}$. [Given : $\sqrt{3} = 1.732$]

Space for Rough Work

4. 50 ml of water sample containing temporary hardness only, required 0.1 ml of M/50 HCl for complete neutralisation. Calculate the temporary hardness of water in ppm.
5. Among the following total number of planar and polar molecules is / are
 Cl_2O , $\text{P}(\text{CH}_3)_3$, $\text{N}(\text{CH}_3)_3$, I_2Cl_6 , CH_3 , NCl_3
6. Find the total number of compound(s), which is/are repelled by magnetic field
 K_2O_2 , KO_2 , O_2 , $\text{NO}[\text{BF}_4]$, KCN , Na_2C_2
-

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I : (Maximum Marks: 24)

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1. Let $f(x) = \frac{2x+1}{x}$, $x \neq 0$, where $g(x) = f(f(x)) - \frac{1}{f(x)}$, $x \neq 0, -\frac{1}{2}$, then identify the correct statement(s) -
 (A) $g(x)$ is an odd function (B) $g(x)$ is an even function
 (C) $g(x)$ is neither even nor odd (D) $g(x)$ is a bounded function
2. The values of 'x' satisfying the equation $\sin^{-1}(x^2 - 5x + 7) = 2\tan^{-1}1$ is (are)
 (A) 2 (B) 3 (C) 4 (D) 5
3. Consider the equation, $\tan^{-1}(4\{x\}) = \tan^{-1}(x + [x])$
 where $[]$ and $\{ \}$ denotes greatest integer function and fractional part function respectively .
 (A) Number of solutions of the equation are 2.
 (B) Sum of all real solution(s) of the equation is integer
 (C) The equation has no real solution
 (D) The product of all real solution(s) of the equation is non zero.

Space for Rough Work

4. Let a, b, c are non-zero real numbers such that $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in arithmetic progression and $a, b, -2c$ are in geometric progression, then which of the following statement(s) is (are) must be true ?
- (A) $a^2, b^2, 4c^2$ are in geometric progression (B) $-2a, b, -2c$ are in arithmetic progression
- (C) $a^3 + b^3 + c^3 - 3abc = 0$ (D) a^2, b^2, c^2 are in harmonic progression
5. Consider a function $f(x) = \frac{|x+1|}{x+1} + \frac{x-1}{|x-1|}$, then choose correct option(s) regarding function f
- (A) many one (B) odd function
- (C) bounded function (D) range contain exactly three integers
6. Consider the series $S_n = 100 + 95 + 90 + \dots + T_n$, then correct statement/s is/are (where T_n represents n^{th} term) -
- (A) $T_{16} + T_{26} = 0$ (B) $T_{11} + T_{31} = 0$ (C) $S_{30} - S_{11} = 0$ (D) $S_{15} - S_{27} = 0$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. Let $f : \mathbb{R} - \left\{ \frac{1}{2} \right\} \rightarrow \mathbb{R} - \left\{ \frac{1}{2} \right\}$, $f(x) = \frac{x-2}{2x-1}$ be a function such that $x = m$ is the solution of $f(x) + 2f^{-1}(x) + 2 = f(f(x))$, then $\frac{m}{4}$ is equal to
2. The number of integers in the domain of the function $f(x) = e^{\ln(\sin^{-1}x)} + e^{\ln(\cos^{-1}x)}$ is
3. If the least value of $f(x) = (\tan^{-1}x)^2 + (\cot^{-1}x)^2$ where a and b are coprime is $\frac{a\pi^2}{b}$, then find the value of $|a-b|$?

Space for Rough Work

4. If $2\sin^{-1}\left(\frac{9-4x^2}{9+4x^2}\right) + 3\cos^{-1}\left(\frac{12x}{9+4x^2}\right) + \tan^{-1}\left(\frac{12x}{9-4x^2}\right) = \lambda\frac{\pi}{2}$ for $x = \frac{-17}{13}$, then $\frac{\lambda}{2}$ is
5. Sum of first three terms of G. P. with non-integer common ratio is 21 and sum of their squares is 189. If S denotes sum of infinite term of this series, then $\frac{S}{4}$ is equal to
6. Minimum possible integral value of the expression $\frac{2a^2\sin^4x + (a^4 + 1)\sin^2x + 3a^2}{a^2\sin^2x}$ (wherever defined) is
-

Space for Rough Work

SECTION-III : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the correct answer is given.

Zero Marks : 0 If no answer is given.

Negative Marks : -1 In all other cases

-
1. Let $f(x)$ be an even function satisfying $f(x) + f(x+2) = f(x+1) \forall x \in \mathbb{R}$, where $f(-13) = 4$, then $f(-5) + f(1)$ is equal to
 2. If $f(x) = \frac{x+a}{x+b}$ and $f : \mathbb{R} - \{-b\} \rightarrow \mathbb{R} - \{1\}$ is such that $f(x) = f^{-1}(x)$, then value of $|b|$ must be
 3. If α, β ($\alpha > \beta$) are two solutions of equation $\tan^{-1}x + \cot^{-1}(-|x|) = 2\tan^{-1}6x$, then $2\alpha + 3\beta$ is equal to
-

Space for Rough Work

4. If range of the function $f(x) = \tan^{-1}(3x^2 + (k - 1)x + 3)$ is $\left[0, \frac{\pi}{2}\right)$ for $x \in \mathbb{R}$, then find the sum of all possible real value(s) of k ?
5. A sequence start with 7120 as its first term and each subsequent term is the sum of cubes of the digits in the previous number. If N is 102^{th} term of this sequence, then sum of digits of N is
6. Let $a_1 = 2$, $a_{n+1} = a_n^2 - a_n + 1$ for $n \geq 1$. Find $\sum_{n=1}^{\infty} \frac{1}{a_n}$
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Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work